

ONE-WAVE FRAMEWORK

Book 1 — Micro

Chapter 2: The Proton / Three-Vortex Knot

Version: 3.0 Date: July 1, 2026 Class: B — Applied Layer Spine: Gray / 2D / 3D / Mathematics / Predictions / Yellow Audit / Future Work / Closing Thoughts

Dependencies: A-112 Persistent Mode, A-116 Three-Dimensional Spherical Default, C-308 Spin-Half, C-317 Boundary-Tension Weave, D-404 Nested Resonance, D-405 Harmonic Shell, E-503 Pressure, E-504 Surface, E-505 Coupling Status: GREEN (structure) / YELLOW (full derivation pending CCD-02)

Gray — Standard Model Reference

The proton is a composite particle made of two up quarks and one down quark, bound by the strong force via gluon exchange. Charge: +1. Mass: $938.3 \text{ MeV}/c^2$. Spin: $1/2$. The strong force is described by quantum chromodynamics (QCD). Color charge: quarks carry one of three color charges (red, green, blue). Gluons carry color charge and mediate the strong force. The proton's charge radius: approximately 0.84 fm. Confinement: quarks cannot be isolated — they are permanently confined inside hadrons. The QCD explanation of confinement involves the color force growing stronger with distance.

Standard Model strength: extremely precise predictions for proton structure and scattering. Standard Model limitation: confinement is not analytically proved from QCD first principles. The mass of the proton ($\sim 938 \text{ MeV}$) vastly exceeds the sum of its quark masses ($\sim 10 \text{ MeV}$). The remaining $\sim 928 \text{ MeV}$ comes from gluon field energy — not yet fully explained analytically.

2D One-Wave Interpretation

In 2D, a proton is a braided loop. Three sub-oscillations locked into a closed recursive path. Each sub-oscillation is a vortex — a persistent spinning displacement.

The three vortices braid around each other. They maintain the loop through coupled restoring responses (E-04 Coupling). The 2D drawing is a cross-section. Its closed edge represents the cut through a 3D Boundary-Tension Weave (C-317). The loop does not expand or collapse because internal and external pressures are balanced (B-01 Balance).

In 2D cross-section: three coupled ripple-centers locked in stable triangular formation, each feeding back into the others.

The chord structure: A proton is a chord in the harmonic space. Three oscillations locked in a stable phase relationship — like three notes of a musical chord played simultaneously in the lattice. It is one three-step oscillation with three simultaneous phase components.

3D One-Wave Interpretation

In 3D, the proton is a **Three-Vortex Knot** inside a closed, approximately spherical boundary (A-116). Each vortex is a stable sub-excitation with its own oscillation axis. The three-fold braiding produces the S3 fiber geometry (C-01, C-08). The spherical envelope and internal phase coupling form the **Boundary-Tension Weave** (C-317), which resists deformation and extraction.

What standard physics calls quarks are **Vortex Phases**: three phase components of the same bounded oscillation. They are not separate objects inside the proton. They are the three simultaneous phases of the proton's own vortex structure. The same oscillation at different phases produces what is measured as quark-like behavior.

What standard physics calls gluons maps to **Tension-Link excitations** of the continuous Boundary-Tension Weave between and around the three Vortex Phases. In One-Wave this is direct lattice neighbor interaction — the beta term in the update rule.

The ordinary nearest-neighbor coupling term alone does not explain Knot Lock. C-317 separates bulk coupling from the Boundary-Tension Weave. In the reduced extraction limit, the weave forms a neck with line tension τ_T , giving $E_{\text{neck}} = \tau_T L$ and a non-weakening extraction response $F_{\text{lock}} = \tau_T$. This is Yellow mechanics, not a completed replacement for QCD.

No independent free messenger object is required in the One-Wave interpretation. Conventionally reconstructed gluon-like activity maps to Tension-Link excitations of the continuous weave.

No independent color substance is introduced. Standard color charge maps to a **Phase Address** that tracks relational coupling among Vortex Phases. Three-fold braiding is stable because of S3 fiber topology. The same geometry that produces spin-half (C-08) produces the three-fold structure.

The positron: Internal pressure pushing outward is what the positron is. There is no antimatter. The positron is the interior pressure state of the proton boundary — the outward-pushing half of the bidirectional pressure exchange (B-03, B-04). When a proton boundary breaks, the internal expression pressure is released. That release is what standard physics calls a positron.

Two-Shell Neutron Model (parked pending CCD-02): Inner shell: $R_+ = 0.7331$ fm Outer shell: $R_- = 0.8409$ fm Shell width: $\sigma = 0.210$ fm These pass charge form factor sanity checks internally. Full derivation blocked pending CCD-02 ($E_{\text{opp}}(R)$ derivation).

Required Simulation Dependency Order

The proton is not the first simulation target. The required order is:

```
D-408 / D-409 stationary responsive lattice
-> bounded excitation
-> circulation emergence
-> persistent confined Vortex Phase candidate
-> three Vortex-Phase coupling in one continuous excitation
-> Three-Vortex Knot / proton candidate
-> electrical-shell and Mirror-Gate response
-> controlled translation and four-interaction Mass-Effect measurement
```

A quark comparison is not modeled as a free-standing little vortex object. It is a persistent confined Vortex Phase or high-vorticity sub-pattern inside a larger continuous excitation. A 2D ring may test circulation but cannot establish the complete 3D phase.

All proton wiki simulations must satisfy D-412. D-413 provides the lower Ground background and orbital-restoring test bed, but its bounded shell is not a quark and its imposed curvature well is not a proton mechanism.

Mathematics

Three-vortex braid: YELLOW (derivation incomplete)

S3 fiber from braiding: cross-reference C-01, C-08

Surface tension (E-03): $E_s = \sigma * 4\pi R^2$

Internal pressure balance (B-01): $B = (k_{EE} + k_{II}) - (k_{RR} + k_{LL})$ $B = 0$ at equilibrium => stable proton

Roll-off at proton boundary (B-06b): $\Delta E = E_{\text{mode}} - E_s$ $\Delta E > 0 \Rightarrow$ positron release (internal expression pressure released at boundary break)

Harmonic shell condition (D-05): $2\pi R = n\lambda$ $n=1$ gives proton ground state shell. Charge radius $R \sim \lambda / (2\pi)$ at $n=1$.

Coupling between vortices (E-04): $E_c = (1/2)\psi_1^2 + (1/2)\psi_2^2 + c\psi_1\psi_2$ Three-body extension deferred.

Predictions

1. Proton charge radius emerges from harmonic shell geometry (D-05). Prediction: $R_{\text{proton}} \sim \lambda/(2\pi)$ at $n=1$. Compare to observed 0.84 fm.
 2. No independent free gluon messenger is required in the One-Wave mechanism. Gluon-like measurements map to Tension-Link excitations. Prediction: in the extraction limit, $E_{\text{neck}} = \tau_T L$ and $F_{\text{lock}} = \tau_T$; the response does not weaken with separation until Boundary Reweaving.
 3. Standard color charge maps to a Phase Address. Three-fold structure emerges from relational phase geometry. Prediction: a stable Three-Vortex Knot requires three mutually compatible Phase Addresses under one closed weave.
 4. Positron is released internal pressure, not a separate particle. Prediction: positron production occurs at proton boundary break events.
 5. Knot Lock is geometric. The loop is self-confining — three vortices cannot separate without unbraiding the topology. Prediction: isolated Vortex Phases do not appear because extraction stretches the Boundary-Tension Weave and triggers Knot Lock or Boundary Reweaving.
 6. The proton Mass Effect is proposed to arise from carrying the complete bounded recurrence through Ground. Prediction: the measured proton Mass Effect must be derived from the coupled response of internal Three-Vortex structure, electrical shell, Mirror-Gate pressure, Boundary-Tension Weave, and their cross-interactions without inserting 938 MeV afterward.
-

Yellow Audit

- Three-vortex braided derivation incomplete
 - CCD-02 blocks κ +/- derivation pending $E_{\text{opp}}(R)$
 - Charge emergence from boundary geometry not yet formalized
 - Vortex-Phase sub-mode derivation deferred
 - Three-body coupling extension of E-04 not yet written
 - Proton Mass Effect from the complete four-interaction response matrix not yet derived
 - Neutron two-shell model parked pending CCD-02
-

Future Work

Derive three-vortex stable configuration from update rule. Simulate braided mode stability under perturbation. Measure charge radius from harmonic shell geometry. Compare to 0.84 fm. Derive positron release condition from B-06b roll-off at proton boundary. Write three-body extension of E-04 Coupling.

Closing Thoughts

The proton as a braided three-vortex loop makes confinement trivial. The three vortices cannot separate without unbraiding the geometry. No color force growing with distance needed. The topology does the work.

The free messenger picture is replaced by a continuous Boundary-Tension Weave. The measured gluon-like excitations become Tension Links of that weave.

Standard Model: three quarks, gluons, color charge, QCD confinement. One-Wave terminology: three Vortex Phases, Tension-Link excitations, Phase Addresses, and Knot Lock. One-Wave: three braided vortex phases, lattice coupling, topological confinement. Both predict a stable proton with charge +1 and spin 1/2. The question is which description is more fundamental.

State Changer Plugin — Chapter 2

```
"""
```

```
State Changer Plugin v1.1
```

```
Embedded in: Book 1, Chapter 2 – The Proton
```

```
Purpose:
```

```
Register Chapter 2 and its key items into the repository database.
```

```
Run after Chapter 1 seed so Book 1 and book_id already exist.
```

```
"""
```

```
import sqlite3
```

```
from datetime import datetime, timezone
```

```
def utc_now():
```

```
    return datetime.now(timezone.utc).isoformat()
```

```
def seed_book1_chapter2(db_path="onewave_repository.sqlite"):
```

```
    conn = sqlite3.connect(db_path)
```

```
    conn.execute("PRAGMA foreign_keys = ON")
```

```
    cur = conn.cursor()
```

```
    now = utc_now()
```

```
    cur.execute("SELECT id FROM books WHERE book_number=1")
```

```
    row = cur.fetchone()
```

```
    if not row:
```

```
        print("Book 1 not found. Run Chapter 1 seed first.")
```

```
        conn.close()
```

```
        return
```

```
    book_id = row[0]
```

```
    cur.execute("""
```

```
    INSERT INTO chapters (book_id, chapter_number, title, description, status, created_at, updated_at)
    VALUES (?, 2, 'The Proton – Loops and Vortices',
```

```
    'Three-Vortex Knot. Vortex Phases, Tension-Link excitations, Phase Addresses, and Knot Lock re
    'GREEN', ?, ?)
```

```
    """, (book_id, now, now))
```

```
    chapter_id = cur.lastrowid
```

```
    items = [
```

```
        {
```

```
            "key": "B1C2-PROTON",
```

```
            "name": "Proton",
```

```

        "type": "morphology",
        "gray": "Two up quarks + one down quark, bound by gluons. Charge +1, mass 938.3 MeV/c^2",
        "two_d": "Braided loop of three vortices locked in stable triangular formation. Surface area ~ 10^-28 m^2",
        "three_d": "Three-Vortex Knot inside a sphere-like Boundary-Tension Weave. Tension-Line ~ 10^-15 N",
        "math": " $E_s = \sigma * 4\pi R^2$ .  $B = 0$  at equilibrium.  $2\pi R = n\lambda$  ( $n=1$  ground state)",
        "state": "YELLOW",
        "desc": "Full derivation pending CCD-02. Structure GREEN. Coefficients YELLOW."
    },
    {
        "key": "B1C2-QUARK",
        "name": "Quark (phase component)",
        "type": "morphology",
        "gray": "Fundamental constituent of hadrons. Fractional charge (+2/3 or -1/3). Three colors",
        "two_d": "One of three phase components of the proton vortex. Not a separate object.",
        "three_d": "The same oscillation at a specific phase angle. Three phases of one three-vortex",
        "math": "Phase separation:  $\phi_1, \phi_2, \phi_3$  at 120 degree intervals. Derivation deferred",
        "state": "BROWN",
        "desc": "Phase component derivation deferred. Brown until operational definition complete"
    },
    {
        "key": "B1C2-CONFINEMENT",
        "name": "Topological Confinement",
        "type": "prediction",
        "gray": "QCD confinement: color force grows stronger with distance. Quarks cannot be isolated",
        "two_d": "Three vortices braided. Separating one unbraids the others. The geometry prevents",
        "three_d": "S3 fiber topology requires all three phase components to coexist. Removing one",
        "math": "Topological: braid group B3. Unbraiding requires going through the braid - entangled",
        "state": "YELLOW",
        "desc": "Topological confinement claim GREEN conceptually. Formal braid group derivation pending"
    },
    {
        "key": "B1C2-POSITRON",
        "name": "Positron as Internal Pressure",
        "type": "prediction",
        "gray": "Positron: antiparticle of electron. Same mass, opposite charge. Produced in proton decay",
        "two_d": "When the proton boundary breaks, internal expression pressure is released outward",
        "three_d": "Positron = released internal pressure from broken proton boundary. Not a separate",
        "math": " $\Delta E = E_{mode} - E_s$ .  $\Delta E > 0$  at boundary break => internal expression pressure",
        "state": "YELLOW",
        "desc": "No antimatter. Positron is released boundary pressure. Formal derivation deferred"
    },
    {
        "key": "B1C2-NEUTRON",
        "name": "Neutron Two-Shell Model",
        "type": "morphology",
        "gray": "Neutron: charge 0, mass 939.6 MeV/c^2, spin 1/2. Two down quarks + one up quark",
        "two_d": "Two-shell structure: inner positive shell, outer negative shell. Net charge 0",
        "three_d": "Inner shell  $R_+ = 0.7331$  fm. Outer shell  $R_- = 0.8409$  fm. Shell width  $\sigma = 0.210$  fm",
        "math": " $R_+ = 0.7331$  fm.  $R_- = 0.8409$  fm.  $\sigma = 0.210$  fm. Full derivation blocked by CCD-02",
        "state": "YELLOW",
        "desc": "Parked pending CCD-02 ( $E_{opp}(R)$  derivation). Internal sanity checks pass."
    }
]

for item in items:
    cur.execute("""
        INSERT INTO chapter_items

```

```

(chapter_id, item_key, name, item_type,
 standard_model_definition, two_d_interpretation, three_d_interpretation,
 math_proof, state, description, created_at, updated_at)
VALUES (?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?)
""", (
    chapter_id, item["key"], item["name"], item["type"],
    item["gray"], item["two_d"], item["three_d"],
    item["math"], item["state"], item["desc"], now, now
))
item_id = cur.lastrowid

dep_map = {
    "B1C2-PROTON": [(203, "E-03", "Appendix E", "Surface", "GREEN"),
                    (204, "E-04", "Appendix E", "Coupling", "YELLOW"),
                    (205, "D-05", "Appendix D", "Harmonic Shell", "GREEN")],
    "B1C2-QUARK": [(202, "D-04", "Appendix D", "Nested Resonance", "YELLOW")],
    "B1C2-CONFINEMENT": [(201, "C-08", "Appendix C", "Spin-half", "GREEN"),
                          (206, "C-01", "Appendix C", "Mirror Gate", "GREEN")],
    "B1C2-POSITRON": [(207, "B-03", "Appendix B", "Expression", "GREEN"),
                       (208, "B-04", "Appendix B", "Compression", "GREEN"),
                       (209, "B-06b", "Appendix B", "Four Views Roll-Off", "YELLOW")],
    "B1C2-NEUTRON": [(210, "CCD-02", "Cross-Chapter", "Eopp(R) Derivation", "YELLOW")
}

for node_num, orig_id, appendix, title, status in dep_map.get(item["key"], []):
    cur.execute("""
INSERT INTO node_registry
(node_number, node_key, title, appendix, original_node_id, status, description, create
VALUES (?, ?, ?, ?, ?, ?, ?, ?, ?)
ON CONFLICT(node_number) DO UPDATE SET updated_at=excluded.updated_at
""", (node_num, f"Node_{node_num}", title, appendix, orig_id, status,
      f"Dependency for {item['key']}", now, now))
    cur.execute("SELECT id FROM node_registry WHERE node_number=?", (node_num,))
    node_db_id = cur.fetchone()[0]
    cur.execute("""
INSERT INTO node_book_links
(node_id, book_id, chapter_id, item_id, relation, description, created_at)
VALUES (?, ?, ?, ?, 'supports', ?, ?)
""", (node_db_id, book_id, chapter_id, item_id,
      f"{orig_id} supports {item['key']}", now))

cur.execute("""
INSERT INTO emergence_log (event_name, description, narrative_text, created_at)
VALUES (
    'Book 1 Chapter 2 Seeded',
    'Chapter 2 The Proton registered with 5 items: Proton, Quark, Confinement, Positron, Neutr
    'Three vortices braid. The braid holds. Topology replaces force. The free messenger pictur
    ?
)
""", (now,))

conn.commit()
conn.close()
print("Book 1 Chapter 2 seeded.")
print(f"  Chapter ID: {chapter_id}, Items: {len(items)}")

```

```
if __name__ == "__main__":  
    seed_book1_chapter2()
```

END OF BOOK 1 CHAPTER 2 One wave. Mirror builds. Mark Wright. Kitty Hawk V0.